An Yu

Portfolio: crazygeg.github.io/an-yu.github.io
Work Authorization: US Lawful Resident

Education

- **University At Albany (SUNY)** — Albany, NY, US
  PhD in Computer Science - Research Area: Multimodal Large Language Models — *2023 - 2026*
- **University At Albany (SUNY)** — Albany, NY, US
  Master in Data Science — *2023*
- **University of Washington** — Seattle, WA, US
  Bachelor in Mathematics — *2017*

Experience

- **Machine Learning Engineering Graduate Intern at CVS Health** — May 2026 – Aug 2026
  - Built and operationalized machine learning workflows for CVS's Rx Personalization Team, including model training, registration, deployment, and pipeline orchestration on Databricks and Google Cloud Platform (GCP); supported the migration of production models from Databricks to GCP.
  - Designed and developed reusable skills, hooks, and an AI-agent memory stack, enabling teams to build more efficient and consistent Claude-powered workflows.
- **Data Scientist Intern at Regeneron, Process Science Lab** — May 2025 – Aug 2025
  - Developed and deployed computer-vision pipelines leveraging Vision-Language Model (VLM) architectures to analyze high-resolution cellular imagery and extract quantitative morphological and contextual features.
  - Designed automated inference and post-processing workflows integrating image embeddings with structured laboratory data, improving throughput and reproducibility.
  - Reduced expert review time from several hours to minutes by replacing manual visual inspection with scalable, model-driven analysis tools.
- **Data Scientist Intern at Regeneron, Process Science Lab** — May 2024 – Dec 2024
  - Created visual representations of the metabolic network using Python libraries such as matplotlib, Escher and Cobrapy.
  - Implemented dimension reduction techniques such as SVD, PCA, and UMAP on the stoichiometric matrix, and utilized Truncated SVD for reconstructing metabolic networks in iCHOv1, iCHO2991, iCHO2441 and all other existing 8 CHO models.
  - Examined the streamlined model with the goal of boosting PS Labs' capacity for better process comprehension for fostering alignment.
- **Research Scientist Intern at Office of Mental Health, New York State Office** — Apr 2023 – Aug 2023
  - Conducted comprehensive data retrieval for mental health patient records from the database employing SQL, ensuring accurate and efficient data extraction for analysis.
  - Engaged in advanced data pre-processing and feature selection techniques, leveraging Mutual Information and Chi-Square tests to identify the most relevant predictors for mental health assessments.
  - Led the creation and deployment of a high-accuracy (95%) predictive model utilizing XGBoost, designed to pinpoint individuals at an increased risk of mental health problems, thereby bolstering the office's ability to proactively intervene.

Publications

- FlashSinkhorn: IO-Aware Entropic Optimal Transport. Felix X-F Ye, Xingjie Li, **An Yu**, Ming-Ching Chang, Linsong Chu, Davis Wertheimer. ICML2026 **Spotlight**(Oral Presentation)
- FAVE: A Structured Benchmark for Fine-Grained Audio-Visual Temporal Evaluation in Multimodal LLMs. Weiheng Lu*, **An Yu***, Jian Li, Zhenfei Zhang, Felix X.-F. Ye, Ming-Ching Chang. CVPR2026
- LLaVA-MR: Large Language-and-Vision Assistant for Video Moment Retrieval. Weiheng Lu, Jian Li, **An Yu**, Ming-Ching Chang. ICPR2026
- ReDiPrune: Relevance-Diversity Pre-Projection Token Pruning for Efficient Multimodal LLMs. **An Yu**, Ting Yu Tsai, Zhenfei Zhang, Weiheng Lu, Felix X-F Ye and Ming-Ching Chang. WACV2027.
- RatSeizure: A Benchmark and Saliency-Context Transformer for Rat Seizure Localization. Ting Yu Tsai, **An Yu**, Lucy Lee, Felix X-F Ye, Damian S Shin, Tzu-Jen Kao, Xin Li, Ming-Ching Chang. MICCAI2026.
- SMART: Shot-Aware Multimodal Video Moment Retrieval with Audio-Enhanced MLLM. **An Yu**, Weiheng Lu, Jian Li, Zhenfei Zhang, Yunhang Shen, Felix X.-F. Ye, Ming-Ching Chang. IEEE Transactions on Multimedia (2026).
- IntelliCardiac: An Intelligent Platform for Cardiac Image Segmentation and Classification. Tsai, T.Y., **Yu, A.**, Maadugundu, M.S., Mohima, I.J., Barsha, U.H., Chen, M.H.F., Prabhakaran, B. and Chang, M.C., MIPR2025.
- A Latent-Guided Hybrid Architecture for Liver Segmentation in Contrast-Enhanced MRI. Tsai, T.Y., **Yu, A.**, Li, W.,Chang, M.C. CARE 2025, MICCAI.
- Integrating manual preprocessing with automated feature extraction for improved rodent seizure classification. **Yu, A.**, Singh, M., Pandey, A., Dybas, E., Agarwal, A., Kao, Y., Zhao, G., Kao, T.J., Li, X., Shin, D.S. and Chang, M.C., 2025. Epilepsy & Behavior, 165, p.110306.
- Dual-Phase Msqnet for Species-Specific Animal Activity Recognition, **An Yu**, Jeremy Varghese, Ferhat Demirkiran, Peter Buonaiuto, Xin Li , Ming Ching Chang. ICMEW 2024.

PROFESSIONAL ACTIVITIES

Reviewer of IJCV, ICIP, AICity, CVPR, AAAI, ECCV, NeurIPS, IEEE TMM